Heat-engine efficiency

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “efficiency” and “cycle” to explain your answer. Sentence starters are optional.

efficiency: useful output divided by input

cycle: a process that returns a system to its initial state

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes **9** times as large. **Compare rms speed**.

Word help: particle = a small constituent, such as an atom or molecule

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = \mathbf{1} - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

Worked reasoning: A heat engine takes **100 J** from a hot source, does **25 J** of work and rejects **75 J**. Efficiency=**25/100**=**25%**.

Your turn: Draw three labeled energy arrows and **check conservation**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Draw and label all energy transfers across the boundary.

Chosen system

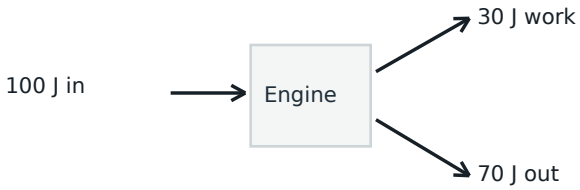
Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An engine receives **100 J**, produces **30 J** of work and rejects **70 J**. **Check the energy balance** and **find efficiency**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A heat engine operates between reservoirs at **600 K** and **300 K**. It receives **800 J** per cycle. **Find the ideal maximum efficiency** and work. Its measured work is **240 J**; **find actual efficiency** and **complete the actual energy account**.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why should a **95%** efficient electric motor not be confused with a **95%** heat engine converting heat from one source?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed speed, mass becomes **4** times as large while energy-coupling fraction becomes **1/6** as large. **Compare coupled energy**.
Show the energy changes and transfers; **label units**.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A heat engine has Q_{hot}=**1500 J** and Q_{cold}=**1050 J**.
1. **Find work** and efficiency.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

An ideal upper bound is η=**1**-T_c/T_h with kelvin temperatures. For T_h=**600 K** and T_c=**300 K**, compare that bound with an engine producing **400 J** work from **1000 J** input. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3